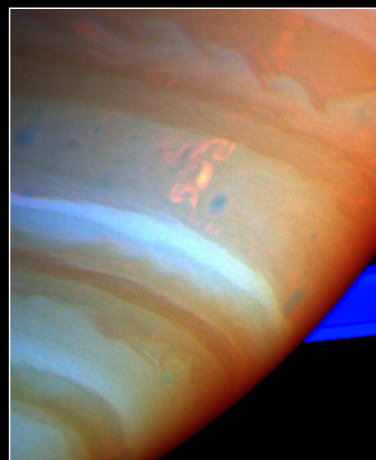


of the cloud layer forms the planet's visible surface. Muted dark belts and bright zones in various shades of yellow girdle Saturn. The muted appearance is due to a thin layer of smoggy haze covering the entire planet. Saturn may appear serene, but things are far from calm. Giant storms in the upper atmosphere

are created by winds generated by heat from within the planet and the planet's rotation. Stormy weather leads to the regular appearances of spots and ribbonlike features.

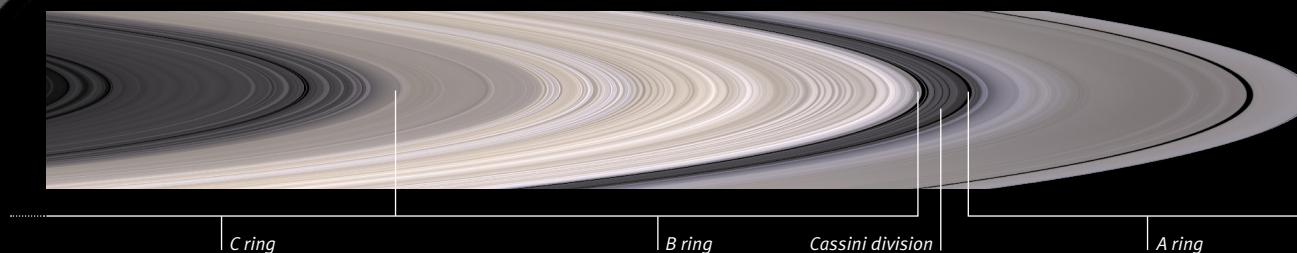
MOONS AND RINGS

Most of Saturn's moons are small and irregularly shaped. Only Titan is larger than our Moon. Most were discovered in the past 25 years, and more are expected. The moons are mixes of rock and water-ice in varying proportions. Some moons orbit within the ring system, which is not solid but is made of billions of pieces of dirty water-ice. These range in size from dust grains to objects several yards across, and they move around Saturn on their own orbits. The system extends for hundreds of thousands of miles into space, but it is only a few miles thick.



Dragon storm

In September 2004, an area in the southern hemisphere known for its storms and dubbed "storm alley" was the setting for a giant thunderstorm. Dragon Storm—the reddish feature above the center in this false-color image—seems to be a long-lived storm that periodically flares up.



Main rings

Saturn's readily-seen ring system comprises three distinct rings: A, B, and C. These are made of ringlets, which in turn are made of pieces of dirty water-ice. The Cassini division, seen from Earth as an empty gap, is in fact full of ringlets.

SATURN

OBSERVATION

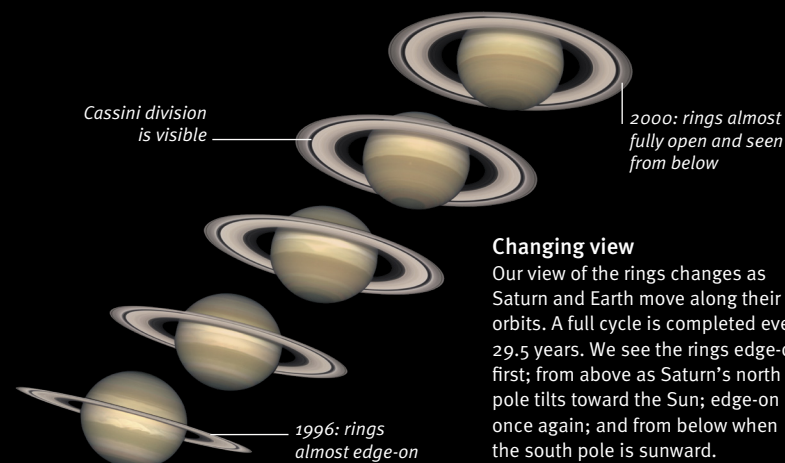
Saturn is twice as remote from Earth as Jupiter but is still bright enough to be seen at some time during the night about ten months of the year. It looks like a star but can be distinguished from its stellar backdrop with the help of the appropriate constellation card.

LOCATING AND LOOKING

Due to Saturn's distance from Earth, it moves more slowly across the sky than closer planets such as Jupiter. Its 29.5-year orbit means it takes about two-and-a-half years to pass through one zodiac constellation. It travels from west to east, but every 12 months it goes into retrograde motion (see p.14), when, for about four months, it appears to move backward.

Like other superior planets, Saturn is best seen when it is on the opposite side of Earth from the Sun (termed "opposition"; see p.14). Its brightness at opposition ranges from 0.8 to a maximum of -0.3. This wide range is largely due to the varying amount of ring facing Earth. Oppositions happen annually, about two weeks later each year. Dates of oppositions are included in the Special Events listings in the Monthly Sky Guide (pp.44-69).

A small telescope or powerful binoculars will reveal the nature of the rings and give a first look at the disk's banded appearance. A larger telescope reveals more disk detail, the three main rings, the Cassini division, and a handful of dots, which are moons. The bulging equatorial region and flattened poles are also apparent. Our view of the rings changes as Saturn orbits the Sun. Their orientation changes from edge-on (when they are virtually invisible), to open (fully visible), and then edge-on again. The rings were last open in 2002. They will be edge-on in 2009, and fully open and seen from above in 2017.



Changing view

Our view of the rings changes as Saturn and Earth move along their orbits. A full cycle is completed every 29.5 years. We see the rings edge-on first; from above as Saturn's north pole tilts toward the Sun; edge-on once again; and from below when the south pole is sunward.



Naked-eye view

Saturn looks like a star to the naked eye, but its warm coloring and disk shape distinguish it from the stars.



Binocular view

Saturn's disk shape is more apparent here. High-powered binoculars will show the rings as bumps on its sides.



Small telescope view

The ring system can be clearly seen, and the planet's bands are just visible.